An Algebra of Prediction-Rewarding Mechanisms

Scoring Rules, Market Makers, and Opinion Pools Generated by One Convex Function

Peter Cotton

Working draft v0.5 · 2026

Abstract

Scoring rules, cost-function market makers, and opinion pools are taught as separate subjects with separate notation. All three are generated by one convex function of the forecast. A proper scoring rule is that function read at the report, a cost-function market maker is its Fenchel conjugate, and the linear and logarithmic opinion pools are the two Kullback-Leibler barycenters. Merging two market makers is infimal convolution, so liquidity adds. The mechanisms are given one input-output signature so that they can be chained, and several familiar statistical methods, Bayesian model averaging, least squares, the Kalman filter and split-conformal prediction, are then chains with one degree of freedom frozen. Every statement is worked on a single three-outcome example.

1 Introduction

Consider an organiser who wants a probability forecast for an event with three outcomes and has three ways to pay for it. She can pay each forecaster by a scoring rule, a fixed formula of the reported probabilities and the outcome. She can open a market in which forecasters buy and sell shares that pay one unit if an outcome occurs, and read the forecast off the prices. Or she can collect the forecasts and average them. The three designs have three literatures.

They are one subject. Fix a convex function G on the set of probability vectors. Reading G and its slope at the reported probabilities gives a proper scoring rule, and every proper scoring rule arises this way (Theorem 1). Taking the convex conjugate of G gives the cost function of an automated market maker, whose prices, freedom from arbitrage, and worst-case loss are three properties of the conjugate (Theorem 2). Averaging forecasts has two natural senses, and they are the two minimisers of a Kullback-Leibler divergence taken in its two directions (Proposition 4).

Results transfer in both directions. The gap between a truthful report and a false one under a proper score is a Bregman divergence, and the same function bounds what the corresponding market maker can lose. Two market makers run side by side behave as one deeper maker, with the depth adding, because conjugation turns a sum of convex functions into an infimal convolution (Proposition 5).

Once the mechanisms are written in one signature, several familiar statistical procedures appear as the same mechanisms with a degree of freedom frozen. Bayesian model averaging is a market of Kelly bettors with entry closed. Weighted least squares is a merge of quadratic makers with their capital fixed. Split-conformal prediction is a two-stage chain whose residual stage admits one unconditional entry, and the freeze forfeits a rent with a closed form, the mutual information between residual and input.

Deployed systems use little of this. Numerai pooled staked submissions into one stake-weighted meta-model and paid each forecast its marginal contribution (Craib et al. 2017). CrunchDAO blends each contest into one ensemble. The IARPA prediction polls aggregated forecasts by track record and compared favourably with head-to-head markets (Atanasov et al. 2017). In every case there is one pool and one internal aggregate, and the mechanism is chosen by convention rather than by the function that generates it.

The mechanisms also chain. The forecast one stage emits can be the message the next stage scores, or the coordinate in which the next stage settles. Chaining requires every mechanism to consume and emit the same kind of object, a distribution, and the mechanisms are written that way. The multi-stage theory and its deployment on the microprediction platform are treated in a companion report (Cotton 2026c). Two related results are treated elsewhere: elicitation from finite point clouds (Cotton 2026e) and the parimutuel account of a conformal predictor's information gap (Cotton 2026a).

The running example has outcomes A, B and C. The forecaster's belief is $q = (0.5, 0.3, 0.2)$, the false report under consideration is $p = (0.7, 0.2, 0.1)$, and the market maker, where there is one, has liquidity $b = 10$. Numbers are rounded to two or three places.

2 Scoring rules and the convex toolkit

The finite setting. Outcomes are a finite set $\{1, \dots, n\}$ and a forecast is a vector p in the probability simplex Δ, with non-negative entries summing to one. In the running example $n = 3$. The continuous setting, densities on $\mathbb{R}$ or $[0,1]^d$, is needed for the opinion pools and the probability integral transform. The finite statements transfer with the usual measure-theoretic care, except where noted.

Scoring rules. A scoring rule pays $S(p, i) \in \mathbb{R} \cup \{-\infty\}$ to a forecaster who reported p when outcome i occurs. Higher is better. A forecaster who believes q and reports p expects to receive $S(p; q) = \sum_i q_i S(p, i)$, assumed well defined with $S(q; q)$ finite. The rule is *proper* if $S(q; q) \geq S(p; q)$ for all p, q: reporting one's belief is never worse than misreporting. It is *strictly proper* if equality forces $p = q$, so that the truth is the only optimal report.

The logarithmic score $S(p, i) = \log p_i$ is the standard example (Good 1952). In the running example the truthful report expects $\sum_i q_i \log q_i = -1.030$ and the false report expects $\sum_i q_i \log p_i = -1.122$. Misreporting costs 0.092 in expectation, and the number is the Kullback-Leibler divergence between q and p.

Convex functions and the Bregman divergence. A function G on Δ is convex when the chord between any two points of its graph lies above the graph. Its slope at p is a vector $G'(p)$ in the subdifferential $\partial G(p)$, the set of slopes of hyperplanes that touch the graph at p and lie below it everywhere. For differentiable G there is one such slope, the gradient. The *Bregman divergence*

$$D_G(q, p) = G(q) - G(p) - \langle G'(p), q - p \rangle$$

is the height of the graph at q above the tangent plane drawn at p. It is non-negative because the tangent plane lies below a convex graph, and it is zero only at $q = p$ when G is strictly convex. For $G(p) = \sum_i p_i \log p_i$ the divergence is the Kullback-Leibler divergence $\mathrm{KL}(q \| p) = \sum_i q_i \log(q_i / p_i)$, which in the running example is 0.092.

The conjugate. The Legendre-Fenchel conjugate of a function R on Δ is

$$R^*(\mathbf{q}) = \sup_{p \in \Delta} \langle p, \mathbf{q} \rangle - R(p), \qquad \mathbf{q} \in \mathbb{R}^n.$$

For each vector $\mathbf{q}$ the conjugate finds the probability vector p that best trades the linear reward $\langle p, \mathbf{q} \rangle$ against the penalty $R(p)$, and records the best value. In the market reading, $\mathbf{q}$ is the vector of outstanding shares and the maximiser is the price vector.

Infimal convolution and conventions. The infimal convolution of f and g is $(f \,\square\, g)(x) = \inf_y f(y) + g(x - y)$, the cheapest way to split x into a part handled by f and a part handled by g. Conjugation turns sums into infimal convolutions, which is the content of Proposition 5. "Closed proper convex" is used in the standard sense (Rockafellar 1970). Differentiable statements are on the relative interior of Δ, with extended-real values allowed on the boundary, and subgradients on Δ act on its tangent space and are defined up to multiples of $\mathbf{1}$.

3 Proper scoring rules are convex functions

The organiser wants the forecaster's belief, not a report tuned to the payment formula. Suppose she pays p_i when outcome i occurs. A forecaster with belief q then expects $\sum_i q_i p_i$, which is largest when all the mass is placed on the most likely outcome. The payment has elicited a mode, not a distribution. Properness rules this out, and Theorem 1 says that properness is convexity.

Theorem 1 (a scoring rule is proper exactly when it is a convex function read at the report; Savage (1971), McCarthy (1956)). *A regular scoring rule S is proper iff there is a convex $G : \Delta \to \mathbb{R}$ with*

$$S(p, i) \;=\; G(p) + \langle\, G'(p),\; e_i - p \,\rangle, \qquad G'(p) \in \partial G(p) \quad (p \in \operatorname{ri} \Delta),$$

where e_i is the i-th unit vector. It is strictly proper iff G is strictly convex relative to Δ, and then $G(p) = S(p; p)$ is the expected score of a truthful forecaster.

The display says that the payment on outcome i is the tangent plane of G at the report, evaluated at the vertex e_i. The expected payment under belief q is therefore the tangent plane evaluated at q, which lies below $G(q)$ by exactly the Bregman divergence $D_G(q, p)$. Truth-telling gains $D_G(q, p) \geq 0$, and strict convexity makes the gain positive for every $p \neq q$.

Proof. ($\Leftarrow$) Since $\sum_i q_i(e_i - p) = q - p$, the expected score is affine in the belief: $S(p; q) = G(p) + \langle G'(p), q - p \rangle$. Hence

$$S(q; q) - S(p; q) \;=\; G(q) - G(p) - \langle G'(p),\, q - p \rangle \;=\; D_G(q, p),$$

which is non-negative by the supporting-hyperplane inequality, and zero only at $q = p$ when G is strictly convex relative to Δ; so S is (strictly) proper.

($\Rightarrow$) Define $G(q) := S(q; q)$. Properness says $G(q) = \sup_p S(p; q)$, and each $q \mapsto S(p; q) = \langle S(p, \cdot), q \rangle$ is affine, so G is a pointwise supremum of affine functions, hence convex. For any fixed p,

$$S(p; q) \;=\; G(p) + \langle S(p, \cdot),\, q - p \rangle \quad \text{with} \quad S(p; q) \leq G(q) \;\; \forall q,$$

so the vector $S(p, \cdot)$ is a subgradient of G at p (acting on the tangent space of Δ, where subgradients are defined up to a constant shift along $\mathbf{1}$, which the representation absorbs). Substituting $G'(p) = S(p, \cdot)$ into the display returns $S(p, i)$ identically. For strictness, note

$D_G(q,p) = 0$ exactly when G has a supporting hyperplane at p that also supports G at q. If G carried a nontrivial affine segment in Δ, a hyperplane supporting it at an interior point of the segment would support it along the whole segment, giving $D_G(q,p) = 0$ for a distinct $q \neq p$, so S would not be strictly proper; conversely strict convexity relative to Δ forces $D_G(q,p) = 0$ only at $q = p$. ■

Theorem 1 is the dictionary *proper scoring rule* ↔ *convex function* ↔ *Bregman divergence* (Gneiting and Raftery 2007; Banerjee et al. 2005). The three classical scores are three choices of G, with the divergences computed directly from the definition:

Score	Generator $G(p) = S(p;p)$	Bregman divergence $D_G(q,p)$
logarithmic	$\sum_i p_i \log p_i$	$\mathrm{KL}(q\|p)$
Brier (quadratic)	$\|p\|_2^2$	$\|q-p\|_2^2$
spherical	$\|p\|_2$	$\|q\|_2\,(1-\cos\theta_{p,q})$

For the logarithmic row the representation gives $S(p,i) = \log p_i$ on the relative interior, with $-\infty$ on the boundary. For the Brier row (Brier 1950) it gives $S(p,i) = 2p_i - \|p\|^2$. For the spherical row, $\nabla G(p) = p/\|p\|$, so $D_G(q,p) = \|q\| - \langle p,q\rangle/\|p\| = \|q\|(1-\cos\theta_{p,q})$, an angular term scaled by $\|q\|$.

In the running example the Brier score pays 0.86, -0.14 and -0.34 on the three outcomes for the false report $p = (0.7, 0.2, 0.1)$. The truthful report expects 0.38 and the false report expects 0.32, so the gap is $0.06 = \|q-p\|^2$. Under the logarithmic score the same misreport cost 0.092. The two rules penalise the same lie by different amounts, and the difference is entirely the choice of G.

4 The same function runs a market

A scoring rule pays one forecaster for one report. A market lets many participants arrive at different times, correct one another, and be paid in proportion to how much each improved the forecast. The automated version is the *cost-function market maker* (Hanson 2003, 2007). The maker sells shares in each outcome, a share paying one unit if that outcome occurs, and quotes the price of any bundle from a fixed convex function C of its outstanding inventory $\mathbf{q}$: a trader who moves the inventory from $\mathbf{q}$ to $\mathbf{q}'$ pays $C(\mathbf{q}') - C(\mathbf{q})$. There is no counterparty search and no empty order book, because the maker always quotes.

Hanson's logarithmic market scoring rule (LMSR) is the case $C(\mathbf{q}) = b \log \sum_i e^{q_i/b}$. In the running example the maker starts with no shares outstanding and quotes each outcome at $1/3$. A trader buys 10 shares of A and pays $C(10,0,0) - C(0,0,0) = 4.53$; the prices move to $(0.576, 0.212, 0.212)$. A second trader buys 5 shares of B for 1.29, and the prices become $(0.506, 0.307, 0.186)$. Had the first trader sold her ten shares straight back, she would have received exactly 4.53: costs telescope along any path of trades, so a round trip costs nothing.

If A occurs after the first trade, the maker has collected 4.53 and pays out 10, a loss of 5.47. Theorem 2 bounds the loss over every possible sequence of trades by $b \log 3 = 10.99$. The maker's capital at risk is set by b, and b also sets how far a trade of a given size moves the price.

The LMSR is the logarithmic score run sequentially. Writing $p_i(\mathbf{q})$ for the price of outcome i, the inventory satisfies $q_i = b \log p_i(\mathbf{q}) + C(\mathbf{q})$, so a trader who moves the prices from p to p' and then sees outcome i nets $b\,(\log p_i' - \log p_i)$. The first trader in the example moved the

price of A from 0.333 to 0.576 and, when A occurs, collects $10\,(\log 0.576 - \log 0.333) = 5.47$. Each participant is paid the improvement in the log score between the prices they found and the prices they left, scaled by b.

Theorem 2 says that every proper scoring rule has such a market, obtained by reading its generator G as a penalty and taking the conjugate. The market's prices, its freedom from arbitrage, and its worst-case loss are then three properties of the conjugate.

Theorem 2 (the conjugate of a proper score is an arbitrage-free market maker with bounded loss; Hanson (2007), Abernethy et al. (2013)). *Let R be closed, proper, strictly convex, and finite-valued on Δ (in particular $R(e_i) < \infty$ at every vertex, which the worst-case-loss bound needs), extended by $+\infty$ off Δ so that $C = R^*$ below is the Fenchel conjugate on $\mathbb{R}^n$, and define*

$$C(\mathbf{q}) = \sup_{p \in \Delta} \left(\langle p, \mathbf{q} \rangle - R(p) \right).$$

Then:

(i) C is convex and finite, the maximiser $p^*(\mathbf{q})$ is unique, and $\nabla C(\mathbf{q}) = p^*(\mathbf{q}) \in \Delta$: prices are a probability vector. (Without strict convexity, prices lie in $\partial C(\mathbf{q})$.)

(ii) C is translation-equivariant, $C(\mathbf{q} + \alpha \mathbf{1}) = C(\mathbf{q}) + \alpha$; costs telescope over any trade path, so round trips cost zero and buying the full bundle costs its payout: the market is arbitrage-free.

(iii) With initial state $\mathbf{q}_0 = \mathbf{0}$, the maker's loss when the market settles on outcome i after net sales $\mathbf{q}$ is

$$\text{loss}_i(\mathbf{q}) = q_i - C(\mathbf{q}) + C(\mathbf{0}) \;\le\; R(e_i) - \inf_{p \in \Delta} R(p) \;\le\; \sup_{p \in \Delta} R(p) - \inf_{p \in \Delta} R(p).$$

(iv) Taking $R(p) = b \sum_i p_i \log p_i$ gives $C(\mathbf{q}) = b \log \sum_i e^{q_i/b}$, Hanson's LMSR, with worst-case loss $b \log n$.

Take $R = G$, the generator of Theorem 1 read as a penalty. Prices are the gradient of the conjugate (Danskin's theorem), no-arbitrage is its translation-equivariance, and the loss bound is the range of the penalty over the simplex (Fenchel-Moreau). For the LMSR the penalty is $b \sum_i p_i \log p_i$, which is 0 at a vertex and $-b \log n$ at the uniform distribution, so the range is $b \log n$.

Proof. (i) C is a supremum of affine functions of $\mathbf{q}$, hence convex; the supremum of a continuous function over the compact Δ is attained, and strict concavity of $p \mapsto \langle p, \mathbf{q} \rangle - R(p)$ makes the maximiser unique. Danskin's theorem gives $\nabla C(\mathbf{q}) = p^*(\mathbf{q})$.

(ii) $\langle p, \mathbf{q} + \alpha \mathbf{1} \rangle = \langle p, \mathbf{q} \rangle + \alpha$ for $p \in \Delta$, so the supremum shifts by α. A trader moving the state $\mathbf{q} \to \mathbf{q}'$ pays $C(\mathbf{q}') - C(\mathbf{q})$ by definition, so costs over any path telescope; a round trip costs zero, and by translation equivariance the bundle $\alpha \mathbf{1}$ costs exactly α, its sure payout.

(iii) The maker collects $C(\mathbf{q}) - C(\mathbf{0})$ and pays q_i, so $\text{loss}_i = q_i - C(\mathbf{q}) + C(\mathbf{0})$. Then $\sup_{\mathbf{q}} \left(\langle e_i, \mathbf{q} \rangle - C(\mathbf{q}) \right) = C^*(e_i) = R^{**}(e_i) = R(e_i)$ (Fenchel-Moreau, R closed), and $C(\mathbf{0}) = \sup_p -R(p) = -\inf_p R(p)$; hence the supremum of the loss over $\mathbf{q}$ is exactly $R(e_i) - \inf_p R(p)$. The final inequality holds because $e_i \in \Delta$.

(iv) Lagrange: maximising $\langle p, \mathbf{q} \rangle - b \sum p_i \log p_i$ subject to $\sum p_i = 1$ gives $q_i - b(\log p_i + 1) = \lambda$, so $p_i \propto e^{q_i/b}$, and substituting back yields $C(\mathbf{q}) = b \log \sum_i e^{q_i/b}$. Then $R(e_i) = 0$ and $\inf R = -b \log n$ at the uniform distribution, so the loss bound is $b \log n$. ■

Decentralised exchanges use a different design, the *constant-function market maker* (CFMM). The pool holds reserves r_1, r_2 of two assets and accepts any trade that leaves an invariant such as $r_1 r_2 = k$ unchanged (Angeris et al. 2021, 2023). The two designs are convertible, though not by the bare conjugacy of Theorem 2.

Proposition 3 (cost-function markets and CFMMs are level-set duals). *Under the monotonicity, concavity, and reserve-domain hypotheses of Frongillo et al. (2024), cost-function prediction markets and constant-function market makers can be converted into one another. In the unconstrained-reserve sign convention a cost function C gives a concave CFMM potential by $\varphi(\mathbf{r}) = -C(-\mathbf{r})$; bounded-reserve versions require a perspective (level-set) construction. The equivalence is a convex level-set duality; it is not the bare Fenchel conjugacy $C \mapsto C^*$.*

Frongillo et al. (2024) give the general equivalence in full, and Angeris et al. (2023) and Angeris et al. (2021) develop the CFMM side.

Example (constant product). Take two outcome tokens with reserves r_1, r_2 and invariant $r_1 r_2 = k$. The pool's portfolio value at prices $(p, 1-p)$ is

$$V(p) = \inf\{p\, r_1 + (1-p)\, r_2 : \; r_1 r_2 = k\} = 2\sqrt{k\, p(1-p)},$$

by the AM-GM inequality, with the infimum attained at $r_1 = \sqrt{k(1-p)/p}$, $r_2 = \sqrt{kp/(1-p)}$. The value is largest, $\sqrt{k}$, when the two outcomes are priced equally, and falls to zero as either price goes to one. V is concave, and reading $R(p) = -V(p)$ as the penalty of Theorem 2 gives a maker whose worst-case loss is the range of R on $[0,1]$, namely $\sqrt{k}$, the geometric mean of the initial reserves. With $k = 100$ and reserves of 10 and 10, the pool is worth 10 at even prices, 6 at prices $(0.9, 0.1)$, and can lose at most 10. The generator is not entropic, which is why constant-product markets and the LMSR price the same claims differently.

5 The two opinion pools

The organiser now holds forecasts $q_1, \dots, q_m$ with weights $w_i \geq 0$, $\sum_i w_i = 1$, from stakes or track records, and wants one forecast. Two averages are in common use. The *linear pool* is $\sum_i w_i q_i$ (Stone 1961; Bates and Granger 1969). The *logarithmic pool* is proportional to $\prod_i q_i^{w_i}$, the weighted geometric mean, renormalised to sum to one (Genest and Zidek 1986).

In the running example take $q_1 = (0.5, 0.3, 0.2)$, $q_2 = (0.2, 0.3, 0.5)$ and equal weights. The linear pool is $(0.35, 0.30, 0.35)$ and the logarithmic pool is $(0.339, 0.322, 0.339)$. Now replace q_2 by $(0, 0.5, 0.5)$, a forecaster who rules A out. The linear pool becomes $(0.25, 0.40, 0.35)$, and the logarithmic pool becomes $(0, 0.551, 0.449)$. The linear pool keeps any outcome that either forecaster finds likely; the logarithmic pool vetoes any outcome that either forecaster rules out.

The two pools are the two senses of "closest to all the forecasts" under the Kullback-Leibler divergence, which is not symmetric. The linear pool minimises the divergence from each forecast to the average. The logarithmic pool minimises the divergence from the average to each forecast.

Proposition 4 (the linear and logarithmic pools are the two KL barycenters). *Let $q_1, \dots, q_m$ be densities with respect to a common dominating measure, $w_i \geq 0$, $\sum w_i = 1$, and for the second display assume $0 < \int \prod_i q_i^{w_i} < \infty$. Then*

$$\arg\min_p \textstyle\sum_i w_i \, \mathrm{KL}(q_i \| p) = \sum_i w_i q_i, \qquad \arg\min_p \textstyle\sum_i w_i \, \mathrm{KL}(p \| q_i) \;\propto\; \prod_i q_i^{w_i}.$$

Proof. For the first, $\sum_i w_i \,\mathrm{KL}(q_i\|p) = \text{const} - \int (\sum_i w_i q_i)\log p$, and $\int \bar{q}\log p$ is maximized over densities at $p = \bar{q}$ (Gibbs). For the second, $\sum_i w_i \,\mathrm{KL}(p\|q_i) = \int p\log p - \int p\,\overline{\log q}$ with $\overline{\log q} = \sum_i w_i \log q_i$; the Lagrange condition is $\log p = \overline{\log q} + \text{const}$. ■

Bayesian model averaging trains the weights by cumulative log score and then mixes linearly, $\sum_m w_m F_m$: a linear pool with score-trained weights. The logarithmic pool multiplies densities and renormalises. They are different aggregates with different sharpness (Genest and Zidek 1986).

6 Merging market makers

Suppose two LMSR makers quote the running example side by side, one with $b = 10$ and one with $b = 5$, and a trader wants 10 shares of A. She splits the order to minimise the total cost, sending 6.67 shares to the deeper maker and 3.33 to the shallower one, in proportion to b, and pays 4.12. A single LMSR with $b = 15$ charges 4.12 for the same ten shares. Two venues, optimally routed, are one venue with the liquidity added.

The reason is that the optimal split is an infimal convolution of the two cost functions, and conjugation turns the infimal convolution of the costs into the sum of the penalties.

Proposition 5 (merging makers is infimal convolution; Rockafellar (1970), Bhaskara et al. (2023)). *For closed proper convex f, g the conjugate identity $(f \,\square\, g)^* = f^* + g^*$ holds unconditionally; recovering the primal form $f \,\square\, g = (f^* + g^*)^*$ needs $f \,\square\, g$ closed and the infimal convolution exact (automatic for the finite-valued penalties here, since $\operatorname{ri}\operatorname{dom} R_1 \cap \operatorname{ri}\operatorname{dom} R_2 = \operatorname{ri}\Delta \neq \emptyset$). Consequently, merging two cost-function makers with penalties R_1, R_2 (cost functions $C_i = R_i^*$) yields the maker with penalty $R_1 + R_2$, and merging LMSR_{b_1} with LMSR_{b_2} yields $\mathrm{LMSR}_{b_1+b_2}$: liquidity adds.*

Proof. $(f \,\square\, g)^*(p) = \sup_x \langle p, x\rangle - \inf_y\{f(y) + g(x-y)\} = \sup_{y,z}\langle p, y\rangle - f(y) + \langle p, z\rangle - g(z) = f^*(p) + g^*(p)$. For the makers, $C_1 \,\square\, C_2 = (R_1 + R_2)^*$ by biconjugacy, and $b_1 G + b_2 G = (b_1 + b_2)G$ for the entropic generator. ■

The same identity is the composition law of the risk-sharing literature. The aggregate of several agents' convex risk measures is their infimal convolution, with the Pareto allocation as minimiser and the common subgradient as the clearing price (Barrieu and El Karoui 2005; Jouini et al. 2008). The market-liquidity reading is developed by Bhaskara et al. (2023).

7 Chaining mechanisms

The common signature. To chain mechanisms, each must accept what the previous one produced. A *transducer* (Mealy machine) is a state machine that reads an input stream and writes an output stream: a tuple $(S, A, B, \delta, \lambda)$ with state space S, input alphabet A, output alphabet B, transition map $\delta : S \times A \to S$, and output map $\lambda : S \times A \to B$, run as $s_{t+1} = \delta(s_t, a_t)$ and $b_t = \lambda(s_t, a_t)$. The mechanisms above share one instantiation. Let Dist denote the set of distributional beliefs and take S the wealth states, $A = \text{Dist}^m \times \mathcal{X}$ (m participant reports and, where the stage settles, a realised outcome), and $B = \text{Dist} \times \mathbb{R}^m$ (an aggregate belief and transfers).

A *stage* is such a transducer, written

$$M : (\text{Dist}^m,\ w,\ x) \longmapsto (\text{Dist},\ w',\ \pi).$$

A stage takes m forecasts, the current wealths and possibly an outcome, and returns one forecast, the new wealths and the payments. A scoring rule is the transfer component λ of a stage, not itself a map Dist $\to$ Dist. A market maker is a stage whose state is the inventory vector and whose emitted belief is the price. An opinion pool is a stage with no outcome argument and zero transfers. Composition wires the belief output of one stage to the belief inputs of the next while state and transfers thread through.

The operators below act on stages.

Sequentialise. Theorem 2: a proper score run sequentially against a wealth state is a cost-function market maker. The LMSR is the logarithmic score sequentialised.

Pool. A proper score gives a batch elicitation mechanism when reports are scored independently and funded externally: properness is inherited report by report. Parimutuel and budget-balanced versions are a different game, because the pot split couples payoffs through the denominator. The price-taking analysis (Cotton 2026e, sec. 2) gives truthful all-in, a symmetric equilibrium at fractional stakes, and degeneracy as the stake fraction vanishes, and beyond that the equilibrium theory is open.

Ensemble. Proposition 4: the linear and logarithmic pools, applied to the belief outputs of several stages.

Merge. Proposition 5: two makers in parallel are one maker with the penalties added.

Serial composition. A joint model is a factorisation, $P(\text{everything}) = \prod_i P(x_i \mid \text{parents})$, and the serial arrangement runs one venue per factor, each pricing its conditional given what is upstream and settling in cascade. The locality that makes this tradable is Hanson's modularity: in a combinatorial logarithmic market scoring rule a bet on $A \mid B$ moves that conditional and nothing else, uniquely among market scoring rules (Hanson 2003, 2007). Pricing a combinatorial market is probabilistic inference, which is why exact LMSR pricing over a combinatorial space is #P-hard (Chen et al. 2008) and why a deployed combinatorial market ran its updates on the junction-tree algorithm (Sun et al. 2012). Pennock and Wellman (1996) map a Bayes net to an economy with one agent per conditional-probability entry and show that equilibrium prices equal the network's probabilities.

Parallel merge and serial composition are the two operations of a commutative semiring, the setting of the generalised distributive law (Aji and McEliece 2000). Assign each factor the potential $\varphi_i = -\log p_i$. Parallel merge adds potentials, which is the conjugate statement of Proposition 5, and serial composition is the min-plus kernel product $\inf_y[\varphi_1(x,y)+\varphi_2(y,z)]$. A market of optimising traders therefore computes min-plus: a chain of stage potentials prices the Viterbi decoding of the corresponding hidden Markov model. On log-quadratic families partial minimisation equals marginalisation up to a constant, by the Schur-complement identity, so there the alternation of a serial step with a parallel merge reproduces the Kalman filter (Kalman 1960) exactly. Off that family a market prices the max-marginal rather than the marginal.

Transform. Run a stage in a transformed coordinate and map back. Whether propriety survives depends on whether the transformation depends on the forecast.

- *Fixed transformations.* Properness is preserved under any fixed transformation of forecast and outcome, and strict propriety iff the transformation is injective (Allen et al. 2023, Prop. 4; Pic et al. 2025, Prop. 1). The Markov-kernel (stochastic channel) extension, strict propriety iff the channel is injective on laws, is Theorem 2 of the point-cloud paper (Cotton 2026e), whose Theorem 1 also exhibits the canonical failure: a kernel-density smoothing seam whose raw-outcome score elicits a deconvolution.

- *Forecast-dependent transformations.* The probability integral transform below transforms outcomes by the reported F itself, so the fixed-transformation theorems do not apply to it. It is a calibration diagnostic rather than a strictly proper score. The stacked-lottery design (Cotton 2020, slides 29-31) is this operator in practice: percentiles from one game feed the next, and calibration is produced by composing monotone maps contributed by competing algorithms.

The probability integral transform. A forecaster who reports a distribution function F for a continuous outcome X has, if the report is right, made $U = F(X)$ uniform on $(0, 1)$. Suppose instead the report is a standard normal and the truth is normal with standard deviation 2. The event $U < 0.025$, which should have probability 0.025, has probability 0.164, and the middle half of the unit interval receives 0.264 instead of 0.5. A stage downstream can detect the departure without knowing the true law.

Proposition 6 (the probability integral transform; Rosenblatt (1952), Dawid (1984)). *If X has continuous CDF F then $U = F(X) \sim \text{Uniform}(0, 1)$, and $z = \Phi^{-1}(U) \sim N(0, 1)$. In the prequential setting, if F_t is the true conditional law of X_t given the past, then $U_t = F_t(X_t)$ is conditionally uniform and the PIT stream is iid uniform.*

Proof. For continuous F, with the generalized inverse $F^-(u) = \inf\{x : F(x) \geq u\}$, the event $\{F(X) \leq u\}$ differs from $\{X \leq F^-(u)\}$ by a set of probability zero, and $\Pr(X \leq F^-(u)) = F(F^-(u)) = u$ by continuity. The prequential statement applies this conditionally at each t. ■

The converse fails. Reporting the unconditional law F of an iid sequence gives $F(X_t) \sim \text{Uniform}(0, 1)$ even if valuable covariates were ignored, so marginally uniform PITs do not certify an informative forecast. A PIT critic witnesses miscalibration relative to its test class; it complements, and does not replace, a proper score that rewards sharp conditional distributions. Split-conformal prediction depends on this distinction. For discrete forecasts the randomized PIT preserves the exact uniform null, while the mid-PIT is a convenient deterministic diagnostic with a different null distribution.

8 Clamped mechanisms

Between a full ecosystem, in which every stage is a live venue with free entry, and a single fitted model, there is a continuum of *clamped* mechanisms: the same composition with some degree of freedom frozen. Ordinary statistical procedures are at the clamped end. Each applies one clamp, and each clamp forfeits a quantity that can be named.

Freeze entry: Bayesian model averaging. A market of Kelly bettors prices the wealth-weighted average of its participants' beliefs, and the wealth updates are Bayes updates, so the market performs Bayesian model averaging with wealths as posterior weights (Kelly 1956; Breiman 1961; Beygelzimer et al. 2012). In the running example two bettors with equal wealth hold $q_1 = (0.5, 0.3, 0.2)$ and $q_2 = (0.2, 0.3, 0.5)$, and the market quotes their linear pool $(0.35, 0.30, 0.35)$. If A occurs the wealths become 0.714 and 0.286, the posterior weights under an equal prior, and the next quote is $0.714\, q_1 + 0.286\, q_2$. Model averaging over a fixed model list is this market with entry closed, and the clamp forfeits whatever a model outside the list would have earned (Blume and Easley 2006).

Freeze capital: least squares. Precision-weighted estimation of a common mean is a merge of quadratic makers, one per observation, each quoting its value with capital equal to its precision. Two observations, 1.0 with variance 1 and 2.0 with variance $1/4$, merge to the estimate

1.8 with precision 5: precisions add because liquidity adds (Proposition 5). Fixing the capitals at the nominal precisions is the clamp. The market version lets a source that has been wrong lose capital, which is the difference between weighted least squares and its adaptive variants.

Freeze the fee: frictionless aggregation. A maker with no fee earns nothing on uninformed flow and cannot recover adverse-selection losses from volume. Setting the fee to zero turns a venue into an aggregation rule. Letting each maker quote its own fee turns the spread into a price, and the aggregate of fee-bearing makers is a consolidated limit order book (Cotton 2026d, 2026b).

Freeze propagation: the Kalman filter. In the filtering recursion the correction step is a parallel merge, executed by trades against an observation maker, while the prediction step is computed by whoever owns the dynamics. Nobody is paid to move the state forward. Unclamping means posting the transition as a venue with securities on the pair (x_t, x_{t+1}), which requires a settlement rule for that venue, and there is no general theorem that self-interested trading performs the elimination.

Freeze the residual venue: split-conformal prediction. Split-conformal prediction fits a point predictor on one half of the data, computes the absolute residuals on the other half, and reports the prediction plus or minus the 90th percentile of those residuals (Lei et al. 2018). The interval covers 90 percent of new outcomes, averaged over inputs, with no assumption on the predictor. As a chain, this is a base predictor feeding a residual stage whose field is restricted to a single *unconditional* entry: one residual law, applied at every input.

In a log-wealth pool settling on the residual, an entrant who conditions on the input X outearns the unconditional entry at the rate $I(R;X)$, the mutual information between residual and input, which in the Gaussian case is $-\frac{1}{2}\log(1-\rho^2)$ (Cotton 2026a). That rate is the rent the clamp forfeits. Marginal coverage is the break-even statement of the clamped venue: the unconditional entry is priced correctly on average, so it is safe, and the conditional information is unpriced. The gap does not close when the base forecast is calibrated, because marginal correctness is not conditional correctness (Foygel Barber et al. 2021). When the residual stage is run as an actual pool the rent is collected; the mechanism ran in production as the microprediction residual streams and the MidOne contest (Cotton 2026c).

The unclamped end. In a nearest-the-pin parimutuel over a continuum, participants submit clouds of samples, the pot is split in proportion to the density each placed at the realised outcome, and entry is open; this was the reward engine behind montepredict (Cotton 2024). Who enters, how much each stakes, what each conditions on, and how sharp a submission is are all unclamped. Open entry brings its own design problems: scoring a kernel density at the raw outcome elicits a deconvolution of the belief rather than the belief (Cotton 2026e).

Each clamp replaces a price with an assumption:

clamp	assumption replacing a price	forfeited
entry closed	the model list is adequate	what an outside model would earn
capital fixed	nominal precisions are right	the correction from realised performance
fee zero	flow is uninformed	adverse-selection revenue
propagation computed	the dynamics are known	the price of the transition
residual unconditional	covariates carry nothing	$I(R;X)$

Only the last entry has a closed form. Unclamping a stage requires a settlement rule for that stage's securities, and a latent quantity with no settlement functional is not a market.

9 Implementation notes

The mechanisms are implemented against a single interface patterned on the skaters time-series contract (Cotton 2026f), a successor to timemachines (Cotton 2021): every stage consumes and emits the same distributional type and threads its own state, so the stage signature is the forecasting contract with wealth in place of model state. The scoring rules are in `mechanisms/scoring_rules.py`, the LMSR in `mechanisms/lmsr.py`, and the two pools in `mechanisms/aggregation.py`. The chained elicitation-then-calibration pipeline runs in `examples/sim_pipeline.py`. Scores are implemented in loss form, with the sign reversed from the reward form.

References

Abernethy, Jacob, Yiling Chen, and Jennifer Wortman Vaughan. 2013. "Efficient Market Making via Convex Optimization, and a Connection to Online Learning." *ACM Transactions on Economics and Computation* 1 (2): 12:1–39. https://doi.org/10.1145/2465769.2465777.

Aji, Srinivas M., and Robert J. McEliece. 2000. "The Generalized Distributive Law." *IEEE Transactions on Information Theory* 46 (2): 325–43. https://doi.org/10.1109/18.825794.

Allen, Sam, David Ginsbourger, and Johanna Ziegel. 2023. "Evaluating Forecasts for High-Impact Events Using Transformed Kernel Scores." *SIAM/ASA Journal on Uncertainty Quantification* 11 (3): 906–40. https://doi.org/10.1137/22M1532184.

Angeris, Guillermo, Tarun Chitra, Alex Evans, and Matthew Lorig. 2023. "Short Communication: A Primer on Perpetuals." *SIAM Journal on Financial Mathematics* 14 (1): SC17–30. https://doi.org/10.1137/22M1520931.

Angeris, Guillermo, Hsien-Tang Kao, Rei Chiang, Charlie Noyes, and Tarun Chitra. 2021. "An Analysis of Uniswap Markets." *Cryptoeconomic Systems* 1 (1). https://doi.org/10.21428/58320208.c9738e64.

Atanasov, Pavel, Phillip Rescober, Eric Stone, et al. 2017. "Distilling the Wisdom of Crowds: Prediction Markets Vs. Prediction Polls." *Management Science* 63 (3): 691–706. https://doi.org/10.1287/mnsc.2015.2374.

Banerjee, Arindam, Xin Guo, and Hui Wang. 2005. "On the Optimality of Conditional Expectation as a Bregman Predictor." *IEEE Transactions on Information Theory* 51 (7): 2664–69. https://doi.org/10.1109/TIT.2005.850145.

Barrieu, Pauline, and Nicole El Karoui. 2005. "Inf-Convolution of Risk Measures and Optimal Risk Transfer." *Finance and Stochastics* 9 (2): 269–98. https://doi.org/10.1007/s00780-005-0152-0.

Bates, J. M., and C. W. J. Granger. 1969. "The Combination of Forecasts." *Operational Research Quarterly* 20 (4): 451–68. https://doi.org/10.1057/jors.1969.103.

Beygelzimer, Alina, John Langford, and David M. Pennock. 2012. "Learning Performance of Prediction Markets with Kelly Bettors." *Proceedings of the 11th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*. https://arxiv.org/abs/1201.6655.

Bhaskara, Adithya, Rafael Frongillo, Elias Lindgren, and Maneesha Papireddygari. 2023. *A General Theory of Liquidity Provisioning for Prediction Markets*. arXiv:2311.08725. https://arxiv.org/abs/2311.08725.

Blume, Lawrence, and David Easley. 2006. "If You're so Smart, Why Aren't You Rich? Belief Selection in Complete and Incomplete Markets." *Econometrica* 74 (4): 929–66. https://doi.org/10.1111/j.1468-0262.2006.00691.x.

Breiman, Leo. 1961. "Optimal Gambling Systems for Favorable Games." *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability* 1: 65–78.

Brier, Glenn W. 1950. "Verification of Forecasts Expressed in Terms of Probability." *Monthly Weather Review* 78 (1): 1–3. https://doi.org/10.1175/1520-0493(1950)078%3C0001:VOFEIT%3E2.0.CO;2.

Chen, Yiling, Lance Fortnow, Nicolas Lambert, David M. Pennock, and Jennifer Wortman. 2008. "Complexity of Combinatorial Market Makers." *Proceedings of the 9th ACM Conference on Electronic Commerce (EC)*, 190–99. https://doi.org/10.1145/1386790.1386822.

Cotton, Peter. 2020. *The Lottery Paradox: A New Use*. https://www.slideshare.net/slideshow/lottery-paradox-csaildec2020/242173597.

Cotton, Peter. 2021. *Timemachines: Time-Series Prediction with a Uniform Skater Interface*. https://github.com/microprediction/timemachines.

Cotton, Peter. 2024. *Montepredict: A Weekly Monte Carlo Competition in Eleven Dimensions*. https://www.montepredict.com.

Cotton, Peter. 2026a. *Betting Against a Conformal Predictor: A Parimutuel Account of the Information Gap*. https://conformalprediction.net/papers/parimutuel/.

Cotton, Peter. 2026b. *Bounded Market Makers from Convex Duality: Self-Set Fees, Dead Bands, and the Order Book as a Theorem*. https://mechanisms.microprediction.org/papers/bounded-market-makers.pdf.

Cotton, Peter. 2026c. *Multi-Stage Solicitation of Probability Distributions: Experiments, Theory and Perspective on Conformal Prediction*. https://mechanisms.microprediction.org/papers/multi-stage-solicitation.html.

Cotton, Peter. 2026d. *Non-Convex Market Makers: Coherence Without Convexity, Friction, and Parallel Composition*. https://mechanisms.microprediction.org/papers/non-convex-market-makers.html.

Cotton, Peter. 2026e. *Scoring Point-Cloud Distributional Submissions*. https://mechanisms.microprediction.org/papers/scoring-point-cloud-distributional-submissions.html.

Cotton, Peter. 2026f. *Skaters: Fast Online Time-Series Models with Composable Transforms*. https://github.com/microprediction/skaters.

Craib, Richard, Geoffrey Bradway, Xander Dunn, and Joey Krug. 2017. *Numeraire: A Cryptographic Token for Coordinating Machine Intelligence*. https://numer.ai/whitepaper.pdf.

Dawid, A. P. 1984. "Present Position and Potential Developments: Some Personal Views. Statistical Theory: The Prequential Approach." *Journal of the Royal Statistical Society. Series A (General)* 147 (2): 278–92. https://doi.org/10.2307/2981683.

Foygel Barber, Rina, Emmanuel J. Candès, Aaditya Ramdas, and Ryan J. Tibshirani. 2021. "The Limits of Distribution-Free Conditional Predictive Inference." *Information and Inference* 10 (2): 455–82. https://doi.org/10.1093/imaiai/iaaa017.

Frongillo, Rafael, Maneesha Papireddygari, and Bo Waggoner. 2024. "An Axiomatic Characterization of CFMMs and Equivalence to Prediction Markets." *15th Innovations in Theoretical Computer Science Conference (ITCS)*, LIPIcs, vol. 287: 51:1–21. https://doi.org/10.4230/LIPIcs.ITCS.2024.51.

Genest, Christian, and James V. Zidek. 1986. "Combining Probability Distributions: A Critique and an Annotated Bibliography." *Statistical Science* 1 (1): 114–35. https://doi.org/10.1214/ss/1177013825.

Gneiting, Tilmann, and Adrian E. Raftery. 2007. "Strictly Proper Scoring Rules, Prediction, and Estimation." *Journal of the American Statistical Association* 102 (477): 359–78. https://doi.org/10.1198/016214506000001437.

Good, I. J. 1952. "Rational Decisions." *Journal of the Royal Statistical Society. Series B (Methodological)* 14 (1): 107–14. https://doi.org/10.1111/j.2517-6161.1952.tb00104.x.

Hanson, Robin. 2003. "Combinatorial Information Market Design." *Information Systems Frontiers* 5 (1): 107–19. https://doi.org/10.1023/A:1022058209073.

Hanson, Robin. 2007. "Logarithmic Market Scoring Rules for Modular Combinatorial Information Aggregation." *The Journal of Prediction Markets* 1 (1): 3–15. https://doi.org/10.5750/jpm.v1i1.417.

Jouini, Elyes, Walter Schachermayer, and Nizar Touzi. 2008. "Optimal Risk Sharing for Law Invariant Monetary Utility Functions." *Mathematical Finance* 18 (2): 269–92. https://doi.or

g/10.1111/j.1467-9965.2007.00332.x.

Kalman, Rudolf E. 1960. "A New Approach to Linear Filtering and Prediction Problems." *Journal of Basic Engineering* 82 (1): 35–45. https://doi.org/10.1115/1.3662552.

Kelly, Jr., J. L. 1956. "A New Interpretation of Information Rate." *The Bell System Technical Journal* 35 (4): 917–26. https://doi.org/10.1002/j.1538-7305.1956.tb03809.x.

Lei, Jing, Max G'Sell, Alessandro Rinaldo, Ryan J. Tibshirani, and Larry Wasserman. 2018. "Distribution-Free Predictive Inference for Regression." *Journal of the American Statistical Association* 113 (523): 1094–111. https://doi.org/10.1080/01621459.2017.1307116.

McCarthy, John. 1956. "Measures of the Value of Information." *Proceedings of the National Academy of Sciences* 42 (9): 654–55. https://doi.org/10.1073/pnas.42.9.654.

Pennock, David M., and Michael P. Wellman. 1996. "Toward a Market Model for Bayesian Inference." *Proceedings of the 12th Conference on Uncertainty in Artificial Intelligence (UAI)*, 405–13. https://arxiv.org/abs/1302.3593.

Pic, Romain, Clément Dombry, Philippe Naveau, and Maxime Taillardat. 2025. "Proper Scoring Rules for Multivariate Probabilistic Forecasts Based on Aggregation and Transformation." *Advances in Statistical Climatology, Meteorology and Oceanography* 11 (1): 23–58. https://doi.org/10.5194/ascmo-11-23-2025.

Rockafellar, R. Tyrrell. 1970. *Convex Analysis*. Princeton Mathematical Series 28. Princeton University Press. https://doi.org/10.1515/9781400873173.

Rosenblatt, Murray. 1952. "Remarks on a Multivariate Transformation." *The Annals of Mathematical Statistics* 23 (3): 470–72. https://doi.org/10.1214/aoms/1177729394.

Savage, Leonard J. 1971. "Elicitation of Personal Probabilities and Expectations." *Journal of the American Statistical Association* 66 (336): 783–801. https://doi.org/10.1080/01621459.1971.10482346.

Stone, M. 1961. "The Opinion Pool." *The Annals of Mathematical Statistics* 32 (4): 1339–42. https://doi.org/10.1214/aoms/1177704873.

Sun, Wei, Robin Hanson, Kathryn B. Laskey, and Charles Twardy. 2012. "Probability and Asset Updating Using Bayesian Networks for Combinatorial Prediction Markets." *Proceedings of the 28th Conference on Uncertainty in Artificial Intelligence (UAI)*, 815–24. https://arxiv.org/abs/1210.4900.